NEP Aligned

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Data Analytics with R

Quality Laboratory Manual

Experiment No. 07

To implement descriptive statistics on a real-world dataset using R

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Experiment No. 07

Title of Experiment: To implement descriptive statistics (mean, median, mode, standard deviation, skewness and kurtosis) on a real-world dataset using R

Aim of Experiment: To compute and analyze descriptive statistical measures such as mean, median, mode, standard deviation, skewness and kurtosis using R on a real-world dataset

System Requirements –

- Operating System: Windows 10 or above
- RAM: Minimum 4 GB
- Hard disk: 10 GB free space
- Processor: 2.33 GHz or higher

Software/s Needed for Experiment

- R programming language (latest stable version)
- RStudio IDE
- Spreadsheet software (Microsoft Excel/ Libre Office)
- Built-in R graphics package
- Internet Connection (for dataset access)/ CSV dataset file

Experiment Objectives:

- To understand the concepts of descriptive statistics
- To compute central tendency measures (mean, median and mode)
- To calculate dispersion using standard deviation
- To analyze data distribution using skewness
- To understand peakedness using kurtosis
- To interpret statistical outputs for real-world datasets

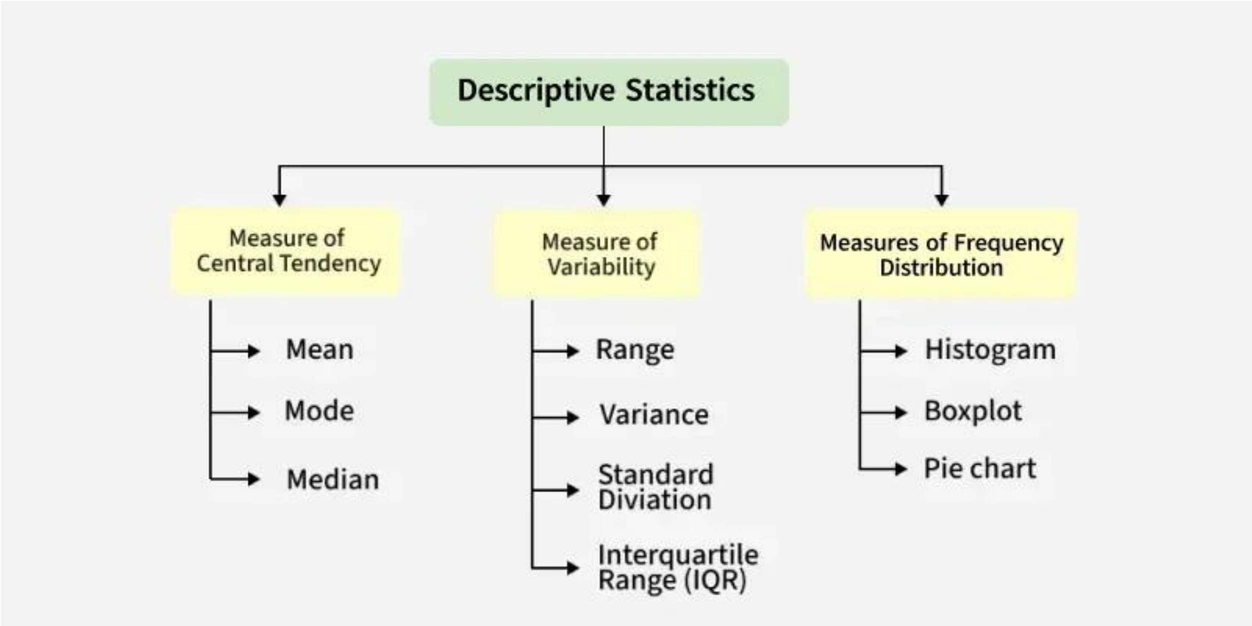
Experiment Outcomes:

After completing the experiment, students will be able to –

- Summarize datasets using statistical measures
- Identify distribution patterns
- Detect variability in data
- Interpret real-world data using statistical metrics
- Apply statistical thinking in data analytics tasks

Theory:

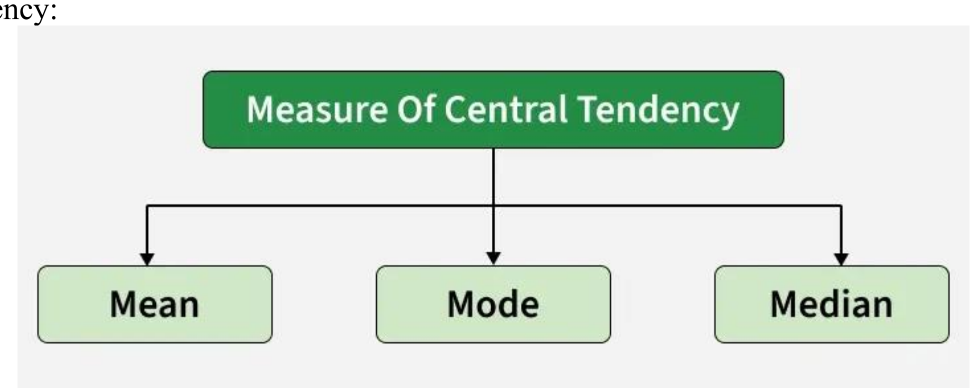
Statistics is the foundation of data science. Descriptive statistics are simple tools that help us understand and summarize data. They show the basic features of a dataset, like the average, highest and lowest values and how spread out the numbers are. It's the first step in making sense of information



Types of Descriptive Statistics
There are three categories for standard classification of descriptive statistics methods, each serving different purposes in summarizing and describing data. They help us understand:

1) Where the data centers (Measures of Central Tendency)
2) How spread out the data is (Measure of Variability)
3) How the data is distributed (Measures of Frequency Distribution)

1. Measures of Central Tendency
Statistical values that describe the central position within a dataset. There are three main measures of central tendency:



Mean: is the sum of observations divided by the total number of observations. It is also defined as average which is the sum divided by count.

$$\bar{x} = \frac{\sum x}{n}$$

Where, x = observations and n = number of terms

Mode: The most frequently occurring value in the dataset. It's useful for categorical data and in cases where knowing the most common choice is crucial.

Median: The median is the middle value in a sorted dataset. If the number of values is odd, it's the center value, if even, it's the average of the two middle values. It's often better than the mean for skewed data.

2. Measure of Variability
Knowing not just where the data centers but also how it spreads out is important. Measures of variability, also called measures of dispersion, help us spot the spread or distribution of observations in a dataset. They identifying outliers, assessing model assumptions and understanding data variability in relation to its mean. The key measures of variability include:

1. Range: It describes the difference between the largest and smallest data point in our data set. The bigger the range, the more the spread of data and vice versa. While easy to compute range is sensitive to outliers. This measure can provide a quick sense of the data spread but should be complemented with other statistics.

Range = Largest data value - smallest data value

2. Variance: It is defined as an average squared deviation from the mean. It is calculated by finding the difference between every data point and the average which is also known as the mean, squaring them, adding all of them and then dividing by the number of data points present in our data set.

$$\sigma^2 = \frac{\sum(x-\mu)^2}{N}$$

Where, x = observation under consideration, N = number of terms, μ = Mean

3. Standard deviation: Standard deviation is widely used to measure the extent of variation or dispersion in data. It's especially important when assessing model performance (e.g., residuals) or comparing datasets with different means.

It is defined as the square root of the variance. It is calculated by finding the mean, then subtracting each number from the mean which is also known as the average and squaring the result. Adding all the values and then dividing by the no. of terms followed by the square root.

$$\sigma = \sqrt{\frac{\sum(x-\mu)^2}{N}}$$

Where, x = observation under consideration, N = number of terms, μ = Mean

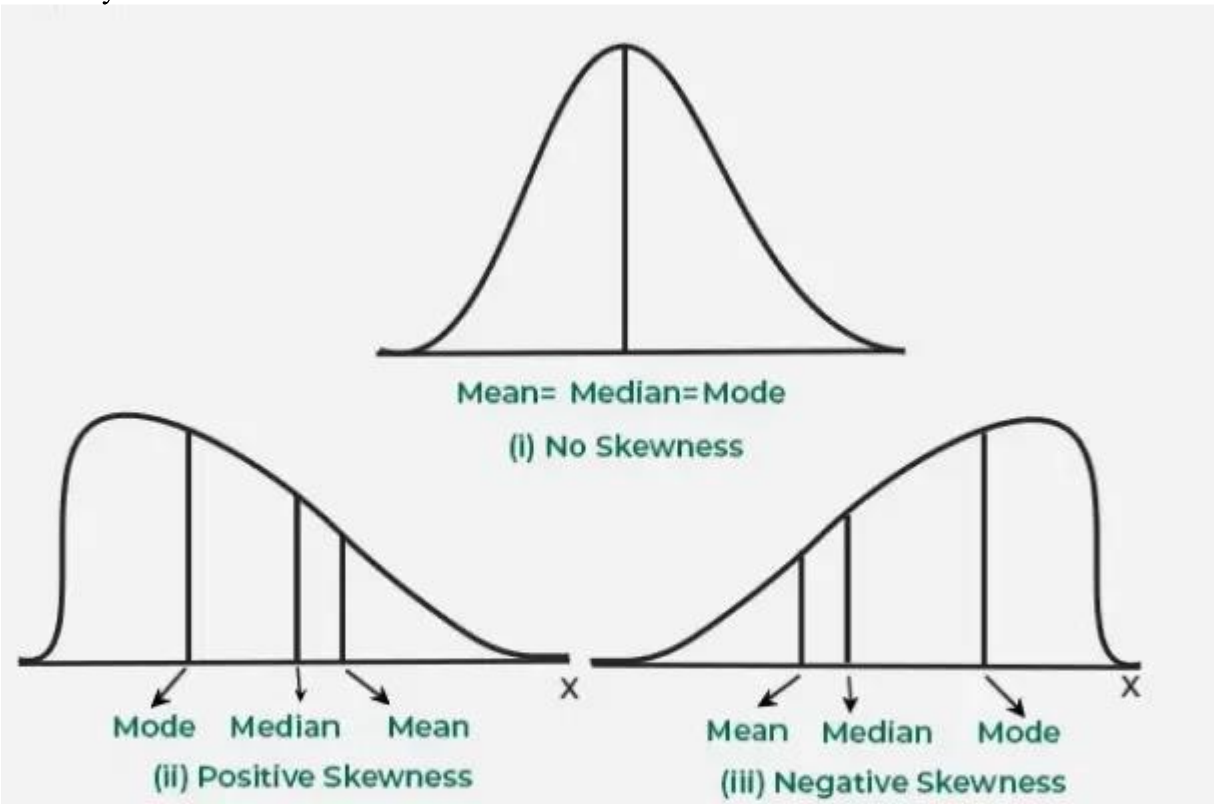
3. Measures of Frequency Distribution
A frequency distribution table is a summarized way to show how data points are distributed across different categories or intervals. It helps to identify patterns, outliers and the overall structure of the dataset. It's often the first step in understanding the dataset before applying more advanced analytical methods or creating visualizations like histograms or pie charts.

Frequency Distribution Table Includes measure like:
- Data intervals or categories
- Frequency counts

- Relative frequencies (percentages)
- Cumulative frequencies when needed

Skewness

Skewness is a key statistical measure that shows how data is spread out in a dataset. It tells us if the data points are skewed to the left (negative skew) or to the right (positive skew) in relation to the mean. It is important because it helps us to understand the shape of the data distribution which is important for accurate data analysis and helps in identifying outliers and finding the best statistical methods to use for analysis



Types of Skewness

Skewness describes the direction and degree of asymmetry in a dataset's distribution. Various types are as follows:

1. Positive Skewness (Right Skew)

In a positively skewed distribution, the right tail is longer than the left which means most data points are on the left with a few large values pulling the distribution to the right.

Relationship:

$$Mean > Median > Mode$$

2. Negative Skewness (Left Skew)

In a negatively skewed distribution, the left tail is longer which means most data points are on the right with a few smaller values pulling the distribution to the left.

Relationship:

$$Mean < Median < Mode$$

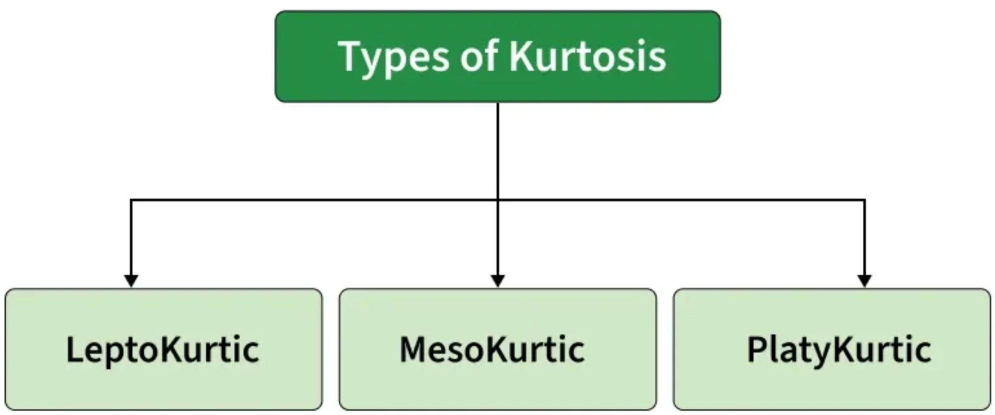
3. Zero Skewness (Symmetrical Distribution)
Zero skewness shows a perfectly symmetrical distribution where the mean, median and mode are equal. In a symmetrical distribution, the data points are evenly distributed around the central point.

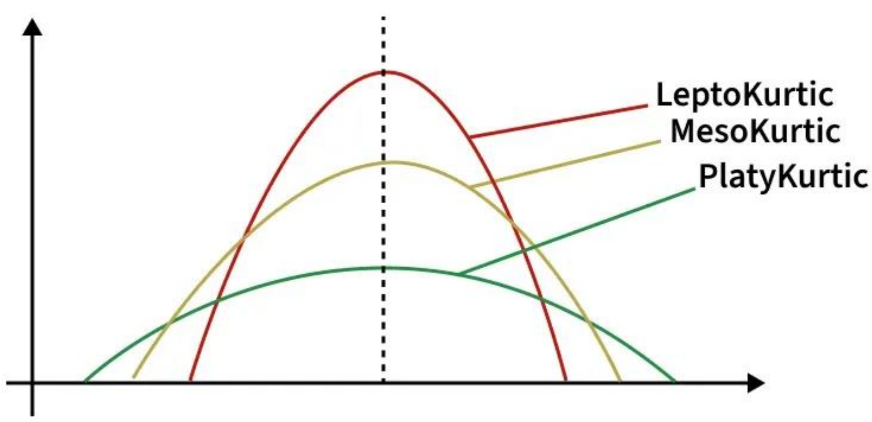
Relationship:

$$Mean = Median = Mode$$

Kurtosis
Kurtosis is a statistical measure that describes the shape of a data distribution especially how heavy or light the tails are. It tells us whether a dataset has outliers than a normal distribution or if most data points stay closer to the average.

It gives analysts a clearer picture of how spread out or peaked the data really is beyond just the mean and variance





Kurtosis quantifies the degree to which data points cluster in the tails or peak of a distribution. The formula is typically based on the fourth standardized moment:

$$Kurtosis = E\,\frac{[(X-\mu)^4]}{\sigma^4}$$

Types of Kurtosis
Kurtosis can be classified as:

1) **Leptokurtic**: Distributions with wide tails and positive kurtosis is called leptokurtic distribution.
2) **Mesokurtic**: When the excess kurtosis is zero or close to zero is called mesokurtic distribution.
3) **Platykurtic**: When the excess kurtosis is negative is called platykurtic distribution.

Task to Perform

Customer spending analysis using descriptive statistics
An e-commerce company wants to analyze the spending behavior of its customers. The management is interested in understanding:

- Average customer spending
- Most common spending pattern
- Variation in spending
- Distribution shape (skewness and kurtosis)

As a data analyst, you are required to compute descriptive statistics using R and provide insights

Dataset –
```
spending = c(
200, 250, 300, 150, 400, 500, 350, 300, 450, 600, 700, 650, 800, 900,
750, 850, 950, 1000, 1100, 1200, 300, 320, 310, 305, 290, 280, 275,
260, 255, 240)
```

Task-1 –
Calculate:

- Mean
- Median
- Mode

Task-2 –
Calculate:

- Standard deviation

Task-3 –
Calculate:

- Skewness
- Kurtosis

Task-4 –
Interpret the results –
- Is the data symmetric or skewed?
- Is the spending distribution normal or not?
- What insights can be derived?

Conclusion:

The experiment demonstrates the computation of descriptive statistical measures using R. students learned how to summarize and interpret datasets using mean, median, mode, standard deviation, skewness and kurtosis. These measures provide essential insights into data distribution and variability, forming the basis for advanced analytics and machine learning applications

Expected Oral Questions:

1) What is descriptive statistics?
2) Difference between mean and median
3) What is mode and where is it used?
4) What does standard deviation represent?
5) What is skewness?
6) What is kurtosis?
7) How do outliers affect mean?
8) Why is median preferred in skewed data?
9) What is normal distribution?
10) Which R package is used for skewness?

FAQs in Interview:

1) Why is mean not reliable in skewed data?
2) What is the difference between variance and standard deviation?
3) What does positive skewness indicate?
4) What does high kurtosis indicate?
5) Why are descriptive statistics important before ML?